Gerrymandering Metrics

2024 Election Results and the 2024 District Map

Metrics

Mean-Median approach:

Remember we need to disregard districts in which there was a choice between two democrats or two republicans !

```
district_level <-  sv_precinct |>
    group_by(DISTRICT)|>
    summarize(DEM01 = sum(DEM01, na.rm= T),
    DEM02 = sum(DEM02, na.rm = T),
    REP02 = sum(REP02, na.rm = TRUE),
    REP01 = sum(REP01, na.rm = T),
    TOTVOTE = sum(TOTVOTE, na.rm =T))|>
        mutate(WINNER= ifelse(DEM01 > REP01, "DEM", "REP"),
        SCALE = (DEM01+DEM02)/TOTVOTE)

# Now we can just compute our score :

district_vi <- district_level |>
    filter(DEM02 == 0 & REP02 == 0)|>
    mutate(v_i = DEM01 / (TOTVOTE))

mean_median <- district_vi |>
    summarize(democratscore = mean(v_i)-median(v_i))

mean_median
```

```
# A tibble: 1 × 1
  democratscore
          <dbl>
1       -0.0125
```

The mean-median score is of -0.00125. But we don't really know what to compare it to! It seems there is a very small negative score in defavor of democrats.

Side quest : ***Is the mean median score statistically significant ?***

```
# We define a function that can compute that systematically on a bootstrap
test:
```

```r
democratscore_fn <- function(data, index) {
  d <- data[index, ]
  mean(d$v_i) - median(d$v_i)
}
set.seed(123)
boot_res <- boot(district_vi, democratscore_fn, R = 2000)

boot.ci(boot_res, type = "perc")
```

```
BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 2000 bootstrap replicates

CALL :
boot.ci(boot.out = boot_res, type = "perc")

Intervals :
Level     Percentile
95%   (-0.0431,  0.0199 )
Calculations and Intervals on Original Scale
```

The CI interval reveals there is no statistical significance on this measure !

Efficiency Score

```r
n_wasted <- function(by, against) {
    w_threshold <- floor((by + against) / 2) + 1
    n_wasted <- ifelse(by > against, by - w_threshold, by)
    n_wasted
}
ES <- district_vi |>
mutate(wasted_D = n_wasted(DEM01, REP01),
      wasted_R = n_wasted(REP01, DEM01))|>
summarize(EG = (sum(wasted_D)+sum(wasted_R))/(sum(TOTVOTE)))

ES
```

```
# A tibble: 1 × 1
     EG
  <dbl>
1 0.500
```

Map !

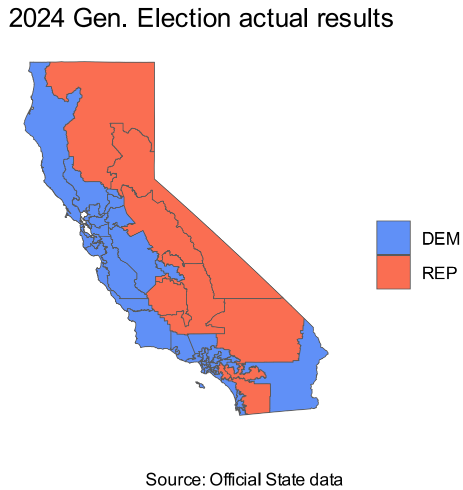
```r
districts <- st_read("/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/Districts/
US_Congressional_Districts.shp")
```

```
Reading layer `US_Congressional_Districts' from data source
  `/Users/sachabondeville/Documents/stat133/Projects/Project-5/gerrymandering-
sachabdvl/data/Districts/US_Congressional_Districts.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 52 features and 4 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -13849210 ymin: 3833656 xmax: -12705030 ymax: 5162404
Projected CRS: WGS 84 / Pseudo-Mercator
```

```r
districts <- districts |>
    mutate(DISTRICT = as.double(GEOID))

districts <- st_sf(left_join(districts, district_level, by = join_by(DISTRICT
== DISTRICT)))
```

```r
districts |>
    ggplot(aes(fill = WINNER))+geom_sf()+
    scale_fill_manual(
    values = c(
      "DEM" = "#6090F7",
      "REP" = "#FA6E52"))+
    theme_minimal(base_family = "Arial")+
    scale_x_continuous(breaks = c())+
    scale_y_continuous(breaks = c())+
    labs(title = "2024 Gen. Election actual results",
    x = "",
    y = "",
    fill = "",
    caption = "Source: Official State data")
```




```
districts <- st_transform(districts, crs = 4326)

#pal_continuous <- colorNumeric(
# palette = colorRampPalette(c("#ff0000ff", "#000dffff"))(100),
#  domain = districts$SCALE
#)

pal_discrete <- colorFactor(
palette = c(
  "DEM" = "#6090F7",
  "REP" = "#FA6E52"
),
domain = districts$WINNER
)
```

2024 Election Results and the proposed 2025 District Map

Methodology :

```
Reading layer `AB604' from data source
  `/Users/sachabondeville/Documents/stat133/Projects/Project-5/gerrymandering-sachabdvl/data/AB604/AB604.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 52 features and 15 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -13857270 ymin: 3832931 xmax: -12705030 ymax: 5162404
Projected CRS: WGS 84 / Pseudo-Mercator
```

We use the registration vote to estimate the amount of DEM votes and REP votes within each block. This estimation will then allow us to study the impact of PROP50 redistribution !

```
block_prop <- sr_precinct |>
    mutate(PROP_DEM_BLOCK = DEM/sum(DEM),
    PROP_REP_BLOCK = REP/(sum(REP)),
    .by = SRPREC_KEY)|>
    mutate(BLOCK_WEIGHTED_DEM = (DEM01+DEM02)*PROP_DEM_BLOCK,
    BLOCK_WEIGHTED_REP = (REP01+REP02)*PROP_REP_BLOCK)
```

Now we have to redistribute these blocks with the new district mapping.

```
hypo_districts <- block_prop |>
    group_by(NEWDISTRICT)|>
    summarize(DEM_est = sum(BLOCK_WEIGHTED_DEM, na.rm = T),
    REP_est = sum(BLOCK_WEIGHTED_REP, na.rm = T))|>
    mutate(WINNER= ifelse(DEM_est > REP_est, "DEM", "REP"))|>
    drop_na(NEWDISTRICT)
```

Metrics :

Mean-Median approach:

Following the same methodology as above :

```
hypo_district_vi <- hypo_districts |>
    mutate(v_i = DEM_est / (REP_est + DEM_est))

hypo_mean_median <- hypo_district_vi |>
    summarize(democratscore = mean(v_i)-median(v_i))

hypo_mean_median
```

```
# A tibble: 1 × 1
  democratscore
          <dbl>
1        0.0276
```

Side quest : *Is the mean median score statistically significant ?*

```
set.seed(43)
boot_res_2 <- boot(hypo_district_vi, democratscore_fn, R = 2000)

boot.ci(boot_res_2, type = "perc")
```

```
BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 2000 bootstrap replicates
```

```
CALL :
boot.ci(boot.out = boot_res_2, type = "perc")

Intervals :
Level     Percentile
95%   (-0.0109,  0.0595 )
Calculations and Intervals on Original Scale
```

The mean-median score increased in favor of the democrats, but the result is still not exactly statistically significant. This reveals the weakness of that measure, as PROP50 *intentionally* redistricted, which should have affected the metric more.

Efficiency Score :

```
hypo_ES <- hypo_district_vi |>
mutate(wasted_D = n_wasted(DEM_est, REP_est),
      wasted_R = n_wasted(REP_est, DEM_est))|>
summarize(EG = (sum(wasted_D)+sum(wasted_R))/(sum(DEM_est + REP_est)))

hypo_ES
```

```
# A tibble: 1 × 1
     EG
  <dbl>
1 0.500
```

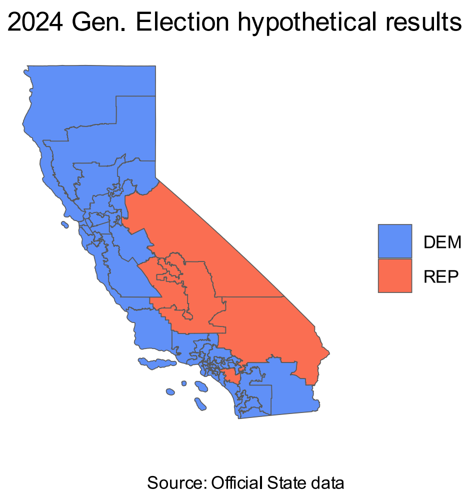
Hypothetical map :

```
hypo_districts <- st_sf(left_join(hypo_districts, ab604, by =
join_by(NEWDISTRICT == DISTRICT)))

hypo_districts <- st_transform(hypo_districts, crs = 4326)


hypo_districts |>
    ggplot(aes(fill = WINNER))+geom_sf()+
    scale_fill_manual(
    values = c(
      "DEM" = "#6090F7",
      "REP" = "#FA6E52"))+
    theme_minimal(base_family = "Arial")+
    scale_x_continuous(breaks = c())+
    scale_y_continuous(breaks = c())+
    labs(title = "2024 Gen. Election hypothetical results",
    x = "",
    y = "",
```

```
    fill = "",
    caption = "Source: Official State data")
```




```
hypo_districts <- st_transform(hypo_districts, crs = 4326)

#pal_continuous <- colorNumeric(
# palette = colorRampPalette(c("#ff0000ff", "#000dffff"))(100),
#  domain = districts$SCALE
#)

pal_discrete_2 <- colorFactor(
palette = c(
  "DEM" = "#6090F7",
  "REP" = "#FA6E52"
),
domain = hypo_districts$WINNER
)
```

Interactive maps code :

```
# Actual results:
leaflet(districts) |>
addProviderTiles("CartoDB.Positron")|>
addPolygons(
fillColor = ~pal_discrete(WINNER),
color = "black",
weight = 1,
fillOpacity = 0.8,
label = ~paste0("DEM Votes:", DEM01, " | REP votes : ", REP01))|>
addLegend(
```

```
pal = pal_discrete,
values = ~WINNER,
title = "Election result",
position = "bottomright"
)

#Hypothetical results:
leaflet(hypo_districts) |>
addProviderTiles("CartoDB.Positron")|>
addPolygons(
fillColor = ~pal_discrete(WINNER),
color = "black",
weight = 1,
fillOpacity = 0.8,
label = ~paste0("DEM Votes:", floor(DEM_est), " | REP votes : ",
floor(REP_est)))|>
addLegend(
pal = pal_discrete_2,
values = ~WINNER,
title = "Election result",
position = "bottomright"
)
```